

Sum and Difference Formulas



Exact values with sum and difference formulas

- 1 Use a sum or difference formula to find the exact value of $\cos 75^\circ$.
 - 2 Find the exact value of $\sin \frac{\pi}{12}$.
 - 3 Find the exact value of $\tan 105^\circ$.
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Working with given values

- 4 Angles A and B are both in Quadrant I with $\sin A = \frac{4}{5}$ and $\cos B = \frac{5}{13}$. Find:
 - a $\sin(A + B)$
 - b $\cos(A + B)$
 - c the quadrant of $A + B$
- 5 Use a sum or difference formula to simplify:
 - a $\sin\left(x + \frac{\pi}{2}\right)$
 - b $\cos(x - \pi)$